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DATE OF SUBMISSION: 5 JUNE, 2026

✓ Question 1: Income Tax Calculator (Basic Operations)

Description

Create a comprehensive income tax calculator that demonstrates all types of Python operators:

- Arithmetic operators (+, -, *, /, //, %,)
- Comparison operators (<, >, <=, >=, ==, !=)
- Logical operators (and, or, not)
- Assignment operators (=, +=, -=, *=, /=)
- Bitwise operators (&, |, ^, ~, <<, >>)
- Membership operators (in, not in)
- Identity operators (is, is not)

Requirements

Create a program that calculates income tax based on the following slabs:

- Income ≤ Rs 2,50,000: No tax
- Income Rs 2,50,001 - Rs 5,00,000: 5% tax
- Income Rs 5,00,001 - Rs 10,00,000: 20% tax
- Income > Rs 10,00,000: 30% tax Use at least one example of each operator type mentioned above in your implementation

1. Display the taxable income, tax amount, and net income after tax
2. Include validation to ensure income is a positive number

Input/Output Example

Enter your annual income: 750000

Taxable Income: Rs 750000

Tax Amount: Rs 62500

Net Income after Tax: Rs 687500

Constraints

- Income must be a positive number

- Handle invalid inputs gracefully

```

try:
    income = float(input("Enter your annual income: "))

    if income <= 0:
        print("Error: Income must be a positive number.")
    else:
        tax_rate = 0
        tax_amount = 0

        if income <= 250000:
            tax_rate = 0

        elif income > 250000 and income <= 500000:
            tax_rate = 0.05

        elif income > 500000 and income <= 1000000:
            tax_rate = 0.20

        else:
            tax_rate = 0.30

        tax_amount = income * tax_rate

        net_income = income - tax_amount

        print("\n----- TAX REPORT -----")
        print(f"Taxable Income: Rs {income:.2f}")
        print(f"Tax Rate: {tax_rate * 100:.0f}%")
        print(f"Tax Amount: Rs {tax_amount:.2f}")
        print(f"Net Income after Tax: Rs {net_income:.2f}")

except ValueError:
    print("Invalid input! Please enter a numeric value.")

```

Enter your annual income: 750000

----- TAX REPORT -----

Taxable Income: Rs 750000.00

Tax Rate: 20%

Tax Amount: Rs 150000.00

Net Income after Tax: Rs 600000.00

Question 2: Student Final Grade Calculator (Global & Local Variables)

Description

Create a student grade calculator that uses both global and local variables to calculate a student's final grade based on multiple assessment components.

Requirements

1. Use a global variable to store the total maximum marks (e.g., 500)
2. Create a function that takes individual subject marks as local variables
3. Calculate the percentage using both global and local variables
4. Determine the grade based on percentage:
 - A: 90% and above
 - B: 80-89%
 - C: 70-79%
 - D: 60-69%
 - F: Below 60%
5. Display student name, total marks, percentage, and final grade

Input/Output Example

Enter student name: John Doe

Enter marks for Math: 85

Enter marks for Science: 90

Enter marks for English: 88

Enter marks for History: 82

Enter marks for Geography: 90

Student: John Doe

Total Marks Obtained: 435

Maximum Marks: 500

Percentage: 87.0%

Final Grade: B

Constraints

- Each subject mark should be between 0 and 100
- Validate all inputs

```
# Global variable  
MAX_MARKS = 500
```

```
def calculate_grade(math, science, english, history, geography):
    # Local variables
    total_marks = math + science + english + history + geography
    percentage = (total_marks / MAX_MARKS) * 100

    if percentage >= 90:
        grade = "A"
    elif percentage >= 80:
        grade = "B"
    elif percentage >= 70:
        grade = "C"
    elif percentage >= 60:
        grade = "D"
    else:
        grade = "F"

    return total_marks, percentage, grade


def get_valid_marks(subject):
    while True:
        try:
            marks = float(input(f"Enter marks for {subject}: "))

            if 0 <= marks <= 100:
                return marks
            else:
                print("Marks must be between 0 and 100.")

        except ValueError:
            print("Invalid input! Please enter a number.")

student_name = input("Enter student name: ")

math = get_valid_marks("Math")
science = get_valid_marks("Science")
english = get_valid_marks("English")
history = get_valid_marks("History")
geography = get_valid_marks("Geography")

total, percentage, grade = calculate_grade(
    math, science, english, history, geography
)

print("\n----- STUDENT REPORT -----")
print("Student:", student_name)
print("Total Marks Obtained:", total)
print("Maximum Marks:", MAX_MARKS)
print(f"Percentage: {percentage:.2f}%")
print("Final Grade:", grade)
```

```
Enter student name: John Doe
Enter marks for Math: 85
Enter marks for Science: 90
Enter marks for English: 88
Enter marks for History: 82
Enter marks for Geography: 90
```

```
----- STUDENT REPORT -----
Student: John Doe
Total Marks Obtained: 435.0
Maximum Marks: 500
Percentage: 87.00%
Final Grade: B
```

> Question 3: String Slicing Operations

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> Question 4: Collection Types - Inventory Management System

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> Question 5 Grade Classification System (Conditions)

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> Question 6 Password Validation System (Iterations)

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> Question 7: Data Processing with Loops (ETL Simulation)

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Question 8: Pattern Matching - Data Format Processor

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> Question 9: String Manipulation - Text Analyzer

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> Question 10: Comprehensive Calculator with All Concepts

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